

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Introduces OptiCrop and its purpose, with navigation to other pages	Frontend (UI)	Done	0.5
2	About Page	Explains project objectives and how ML-based crop prediction works	Frontend (UI)	Done	0.5
3	Crop Prediction Form	Collects soil/climate parameters (N, P, K, temperature, humidity, pH, rainfall) from the user	Frontend (UI)	Done	0.75
4	Data Preprocessing Pipeline	Handles null value checks, outlier treatment (IQR + log transform), and seasonal feature extraction	Backend / Data Processing	Done	0.75
5	ML Model Training	Trains K-Means Clustering and Logistic Regression models on the agricultural dataset	Backend / ML Model	Done	0.75
6	Model Evaluation & Selection	Evaluates models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix; saves best model	Backend / ML Model	Done	0.75
7	Real-Time Crop Prediction	Flask backend passes user input to the trained model and returns a crop recommendation	Backend / Flask Integration	Done	0.75
8	Application Testing	End-to-end testing of Home, About, and FindYourCrop pages for correct navigation and prediction output	Testing / Validation	Done	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Home Page, Prediction Form, Data Preprocessing, Model Training, Model Evaluation, Real-Time Prediction)

Additional / Nice-to-Have Features	2 (About Page, Application Testing)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, About Page, Crop Prediction Form
2	Backend / Logic	3	Data Preprocessing Pipeline, ML Model Training, Model Evaluation & Selection
3	Database / Storage	1	CSV Dataset + Pickle Model File Storage
4	API / Integration	1	Real-Time Crop Prediction (Flask ↔ ML Model Integration)
5	Security / Authentication	0	Not applicable — no authentication implemented in current scope